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Case Study: Securing the university Examination System

A university user an operating system-based examination management system to store student records, examination marks, faculty details & question paper. The system has three main user groups: students, faculty & administration. Students should be able to view only their own marks & download permitted learning material faculty member needs to view student records & enter or modify marks Administrators manage user accounts, permission & system configuration.

The university has assigned permissions through roles but an adult review identifies the following events.

The university has assigned permissions through the role but an audit review identifies the following events.

- At 10:00 the account Student07 has 3 failed log in attempts.
- At 10:03, Student07 successfully logs in.
- At 10:04, Student07 attempts to open Exammarks.xlsx, which contains marks for a entire class.
- At 10:05, the system records an attempted modification of Exammarks.xlsx by Student07
- At 10:15, Faculty02 modifies marks in Exam Marks.xlsx & saves the files successfully
- At 10:20, an administrator discovers that a former faculty member's account is still active
- The audit log is stored on the same server, & user with broad permissions may be able to modify or delete it.

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Q.1. multiple failed login Attempts: student07. has three failed login attempts, which may indicates an attempt to access an account without permission.

unauthorized file Access: student07 tries to open Exammarks.xlsx, which contain the marks of whole class, even though student should only view their own marks

unauthorized modification of marks: student07 modifies the examination, marks file & saves it successfully, showing that the system does not properly restrict access

weak Access control: A former faculty members account is still active, allowing them to access the system even after leaving the university.

Q.2	Authentication	Authorization
a.	Verify the users identity	Checks the user's permission
b.	Asks "who are you?"	Ask "what are you allowed to access?"
c.	Happens before authorization	Happens after authentication

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Q.5 Explain how the principle of least Privilege should be applied in this System. State what each group needs to do & which unnecessary permission should be removed.

Q.6. Interpet the Audit events from 10:00 to 10:05 what should the security team to do next?

Explain why the sequence is suspicious without assuming that is proves an attack.

Q.7. Identify at least five events that should be recorded in the audit logs include login, file, account, permission, & for administrative, events.

Q.8. Explain why the audit logs must be protected & recomend two way to protect them.

Answers

The university wants to protect confidentiality Integrity, & availability; prevent authorized access; maintain accountability; & Investigate suspicious activity. You are asked to recommend appropriate operation system, access-control, security administration & auditing measure.

Q.1. Identify the main security problem in the case study.
list at least four risk or weakness indicated by the scenerio.

Q.2. Explain the difference between authentication & authorization using student07 as an example distinguish identity verification from permission checking.

Q.3. Recommend on, access-control policy for students faculty & Administration. justify your answer. consider how permission should be assigned

Q.4. Prepare a suitable Access control list for Exam marks.xlsx.
specific appropriate Read, write & Delete permission for each user group

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Student07 may be successfully authenticated but they should not be authorized to access or modify the marks of the entire class.

Q.3. The university should use Role-Based Access control (RBAC), where permission are assigned according to each users role

- Students: Can view their own marks & download permitted learning materials.
- Faculty: Can view Student record & enter or modify marks for their assigned classes.
- Administrators: Can manage user accounts, permissions permission & system settings.

& permission Management:

a) Follow the least privilege principle by giving user only the, permission they need.

b) Review permissions regularly & deactive accounts when user leave.

c) maintain secure audit logs to track Important activities.

JUSTIFICATION - RBAC simplifies permission management present unauthorized access & protect confidentiality

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Aat Student records & examination marks.

Q.4. The Rollowing Access control list (ACL) defines permission Ror examinarks.xlsx.

User Group	Read	Delete	write
Students	own marks only	NO	NO
Faculty	Yes, Ror assigned classes	NO	Yes Ror assigned marks.
Administrators	Yes	only iR authorized	only iR authorized
Former Raculty	NO	NO	NO

This Act protects examination marks Rrom unauthorized aecess, modiRication & deletion It ensures that users can access only the Information required Ror their roles.

backups, to prevent data loss or tampering

Q.9) The University should immediately deactivate the former faculty members accounts to prevent unauthorized access

1) Access Removal - Revoke all permission & access to university systems

2) Account Management - Regularly review accounts & deactivate those belonging to formers Employee

3) Security Check - Review recent login activity to identify any suspicious access

Proper account management ensures that only current, authorized users can access university resources & confidential data.

Q.10) The university should follow these steps:

- Investigation - Review audit logs login attempts & changes made to Exammax
- Access Review - Deactivate the former; activity Members account & check permission for all users.
- Corrective Action - Restore the original marks if unauthorized changes are found

investing the events before confirming a security breach.

Q.7]

1. login events: successful & failed login attempt.
2. File events: opening modifying, saving or deleting Exammarks.xlsx
3. Account Event: Creating updating or deactivating users account.
4. Permission event: Granting, changing or removing access permission.
5. Administrative event: Changes to system setting or security policies.

Q.8] Audit logs must be protected to prevent unauthorized access modification or deletion They help the security team investigate incidents & identify suspicious activities.

Two ways to protect audit logs.

1) Restrict access - Allow only authorized security staff to access & manage audit logs.

2) Secure Backup - Store logs on one separate secure server & maintain

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Q.5] • Students: Should view only their own
• Faculty: Should access & update marks only for their assigned classes & Remove Permission to delete files or manage user accounts.
• Administrator: Should manage user accounts & System settings Remove unnecessary permission to modify examination marks.
• Former Faculty: Their accounts Should be deactived Immediately after leaving the University.

Q.6] Audit Events:

a. 10:00 :- Student07 has 3 failed login attempt
b. 10:03:- Student07 successfully logs in
c. 10:04:- Student07 successfully to accesses Exammarks.xlsx
d. 10:05:- Student07 modifies & saves the file.

The repeated login features followed by ~~wrong~~ authorized access & modification OR confidential marks may include suspicious activity but do not prove an attack.

Next Steps:
1. Verify the login activity.
2. Check file changes & restore data if needed
3. Review permission & Audit log.
4. Restrict suspicious access while investigating

& Fix access control weakness.

- Policy changes - Implement role-based access control, least priviledge & secure audit logs.
- Follow up monitoring - Regularly monitor login attempts. File change & user activities to detect suspicious behaviour

These actions will improve security protect examination data & reduce the risk of future incidents.